Hafza Noor

Researcher in Computer Vision, Trustworthy AI & Optimization

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RESEARCH PROFILE

Mathematics researcher working on **trustworthy deep learning for brain tumor MRI**: classification, 3D multimodal segmentation, and the evaluation questions that decide whether a model can be trusted, namely **data leakage, calibration, explainability and out-of-distribution behavior**. Background in optimization theory and metaheuristics, with released code and live demos for each project. Seeking a fully funded PhD position in medical image analysis or trustworthy AI.

RESEARCH INTERESTS

- Medical image analysis and computer vision: deep learning for segmentation and classification
- Trustworthy AI: explainability, calibration, robustness and dataset auditing
- Mathematical optimization and metaheuristic algorithms; optimization-based machine learning

RESEARCH EXPERIENCE

Explainable Brain Tumor MRI Classification with Uncertainty-Aware Decision Support

Completed, 2026

Research project supervised by Dr. Fiza Zafar, CASPAM, BZU Multan

- **Model:** fine-tuned MobileNetV2 in PyTorch for four-class brain tumor MRI classification (glioma, meningioma, pituitary, no tumor) on a public Kaggle dataset; **95.2% accuracy and macro F1 0.951** on 1,600 held-out test images.
- **Data audit:** designed a train–test leakage check using perceptual hashing and pixel correlation; found **582 test images (36.4%) duplicated in training**, including 88% of the no-tumor class. On the 1,018 clean images, accuracy fell to **92.8% (macro F1 0.892)**.
- **Error analysis:** identified glioma as the weakest class (recall 82%), with 32 gliomas misclassified as no tumor, a clinically critical false-negative mode.
- **Calibration:** Expected Calibration Error with reliability diagrams; ECE rose from 0.028 to 0.045 on non-duplicated images, and temperature scaling (T = 1.83) reduced negative log-likelihood by 15%.
- **Explainability and safety:** Grad-CAM attention maps, confidence-based escalation to human review, and retrieval-augmented reports (MiniLM embeddings, FAISS); observed skewed attention maps and low-confidence errors on out-of-distribution scans.
- **Deployment:** interactive applications on Hugging Face Spaces (Gradio) and Streamlit Cloud; evaluation code released on GitHub. **Tools:** Python, PyTorch, Scikit-learn, Grad-CAM, FAISS, Gradio. Live demo | Code & evaluation

Brain Tumor Segmentation in Real-World MRI Data

Ongoing, 2025 – present

Research under Dr. Fiza Zafar, CASPAM, BZU Multan

- **Groundwork on a public benchmark:** built a 3D multimodal segmentation pipeline on the Medical Segmentation Decathlon brain tumour task (484 MRI volumes; FLAIR, T1w, T1gd, T2w) with a patient-level train/validation/test split (338/73/73) to prevent leakage.
- **Methods:** per-modality z-score normalization on brain voxels; whole tumor, tumor core and enhancing tumor region encoding; lightweight 3D U-Net with instance normalization trained with Dice loss on 128^3 patches (PyTorch, Kaggle GPU).
- **Status:** training and per-region Dice evaluation in progress; prototype Gradio interface with textbook retrieval (FAISS). Next step: extension to real-world clinical MRI data. Code

Feature Selection for Tumor Classification using the Artificial Bee Colony Algorithm

Completed, 2025

Undergraduate research project supervised by Dr. Fiza Zafar (grade 4.00/4.00); presented at a departmental seminar

- Formulated feature subset selection on tumor classification data as an optimization problem minimizing classification error, solved with the Artificial Bee Colony swarm-intelligence metaheuristic.
- Compared the selected subsets with traditional feature selection methods on accuracy, convergence behavior and computational cost. **Tools:** Python, NumPy, Pandas, Scikit-learn, Matplotlib. Code

PRESENTATIONS

- *Feature Selection for Tumor Classification using the Artificial Bee Colony Algorithm*, departmental seminar, CASPAM, BZU Multan, 2025.
- *Explainable Brain Tumor MRI Classification with Uncertainty-Aware Decision Support*, graduate class presentation, BZU Multan, 2026.

EDUCATION

MPhil Mathematics, Bahauddin Zakariya University (BZU), Multan Sep 2025 – present (expected 2027)

- **CGPA 3.70/4.00**, 4th in class. Research supervisor: Dr. Fiza Zafar.
- **A grades (4.00/4.00):** Deep Learning with Neural Networks, Optimization for Machine Learning, Integral Inequalities and their Applications.

BS Mathematics, Bahauddin Zakariya University (BZU), Multan Sep 2021 – Jun 2025

- **CGPA 3.25/4.00**, First Division; semester GPA 3.61 in each of the final two semesters. Centre for Advanced Studies in Pure and Applied Mathematics (CASPAM).
- **A grades (4.00/4.00):** Optimization Theory, Final-Year Project, Design and Analysis of Algorithms, Time Series Analysis; Matrix Computation (3.70).
- **Relevant coursework:** Linear Algebra and Applications, Machine Learning I, Numerical Analysis I, Operations Research, Statistics I & II, Real and Functional Analysis, Ordinary and Partial Differential Equations, Discrete Mathematics.

SKILLS

- **Machine Learning & Vision:** CNNs and transfer learning, 3D U-Net segmentation, Dice loss, Grad-CAM, model calibration, out-of-distribution analysis, dataset leakage auditing, regression, classification, clustering, time series (ARIMA, ETS, LSTM), image enhancement, feature matching.
- **Mathematics & Optimization:** linear algebra, matrix computation, optimization theory, linear and non-linear integer programming, network flow optimization, numerical analysis, probability and statistical inference, integral inequalities.
- **Metaheuristics:** Artificial Bee Colony, Particle Swarm Optimization, Genetic Algorithms, Simulated Annealing.
- **Programming & Tools:** Python (PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, nibabel, FAISS, Matplotlib), SQL, Git/GitHub, Gradio, Streamlit, Hugging Face Spaces, Power BI, DAX, Advanced Excel.
- **Languages:** Urdu (native); English (proficient; Medium of Instruction certificate); Chinese (beginner; Chinese Language I, grade A).

SELECTED PROJECTS

- **Image Enhancement and Feature Matching (OpenCV):** histogram equalization, Gaussian filtering, Canny edge detection, ORB keypoints and brute-force matching. Code
- **Gradient Descent from Scratch:** batch gradient descent for salary prediction without ML libraries, with hand-derived gradients and a study of learning-rate effects on convergence.
- **Economic Growth Prediction (Multiple Linear Regression):** GDP modeled from population, GNI, trade and capital formation; multicollinearity checks, residual diagnostics, QQ plots, R^2 and MSE. Code
- **Data Analytics Dashboards (Excel, SQL, Power BI):** sales, trade, resource utilization and education dashboards with multi-source data models, DAX KPIs and Power Query automation. Trade dashboard

TEACHING, PROFESSIONAL EXPERIENCE AND LEADERSHIP

Mathematics Instructor, Gulshan Academy, Multan 2022 – 2024

- Taught FSc mathematics (including Maple) for two years; designed and graded assessments and gave one-on-one support in problem solving and mathematical reasoning.

Freelance Data Analyst & Consultant, Remote Jan 2024 – present

- Built automated data pipelines, cleaned and explored client datasets, and developed Power BI dashboards and data models (Python, SQL, DAX).

Girls Representative, BS Mathematics, BZU Multan 2023 – 2024

- Represented female students of the program and liaised with faculty on academic and administrative matters.

Executive Member, CASPAM Youth Development Council (CAYDEC), BZU Multan 2022 – 2024

- Organized academic and extracurricular events, workshops and seminars, and supported student development initiatives.

CERTIFICATIONS

- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning (DeepLearning.AI, Coursera)
- IBM Data Science Professional Certificate (Coursera); Excel for Data Analysis (IBM); ETL and Data Pipelines with Shell, Airflow and Kafka (IBM)
- Introduction to Structured Query Language (SQL) (University of Michigan); Business Analytics for Decision Making (University of Colorado Boulder)